

ZIYI GUO

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EDUCATION

Wuhan University, China

Sep 2022 - Jun 2026

Hongyi Honor College

Major: Microelectronics Science & Engineering (EE) | GPA: 3.94/4.00

Relevant Coursework:

- *Electronics*: Analog Circuits (95), Solid State Physics (95), Semiconductor Physics (94)
- *Computer Science*: Data Structures (97), Operating Systems (95), AI Fundamentals (95), Computer Architecture
- *Mathematics*: Numerical Linear Algebra (100), Adv. Mathematics A (96)

Honors & Awards:

- **National Scholarship** (Top 0.2% nationwide, 2024)
- Innova International Exchange Scholarship, \$10,000 award (2025)
- University Outstanding Student Scholarship & Merit Student (2023 & 2024)

University of California, Berkeley

Jan 2025 - May 2025

Visiting Student, Berkeley International Study Program

Relevant Coursework: Digital Design and IC Circuits (A)

National University of Singapore

Aug 2026 - Present

1st Year PhD Student in Computer Science

Supervised by Dr. Trevor E. Carlson

PUBLICATIONS

1. X. Zheng[†], **Z. Guo**[†], et al., **A Unified Function Processor with Integer Arithmetic Based on Piecewise Chebyshev Polynomial Approximation**. *Accepted at IEEE International Symposium on Circuits and Systems (ISCAS 2026)*

[†]These authors contributed equally to this work.

RESEARCH EXPERIENCE

Energy-efficient Acceleration System on Chip (SoC) for Generative AI

Jun 2025 - Present

Duke University, Durham, USA

Module leader | Supervised by [Prof. Yiran Chen](#) and [Dr. Changchun Zhou](#)

- Designed a **Unified Function Processor (UFP)** that accelerates nonlinear functions (ReLU, GeLU, Softmax, etc.) using polynomial/LUT fitting, achieving execution speed comparable to linear operations while supporting parallel 32-element processing and multi-bitwidth computation. The design reduced rMAE by > 75% (sigmoid), area by 93.6%, and energy by ~ 79% compared to prior approaches.
- Built a **Sample Processing Module (SMP)** for up/downsampling with compact memory layout, kernel-size independent reuse, and minimal write operations, effectively reducing memory overhead by 80%.
- Implemented a **Data Similarity Module (DSM)** with pipelined cosine similarity and configurable precision to enhance data reuse and reduce computational cost.
- Led module-level design and Verilog implementation, coordinating integration across three IPs to ensure energy-efficient acceleration for generative AI workloads.

- Performed **C–Verilog co-simulation** via DPI-C, enabling system-level verification of SoC and validating end-to-end functionality with C-based reference models.
- Preparing for tape-out in **16 nm CMOS process**; RTL freeze completed, pre-layout verification in progress.

LLM-specialized Accelerator SoC (Generation II)

Feb 2026 - Present

Duke University, Durham, USA

NPU Architecture Lead | Supervised by [Prof. Yiran Chen](#) and [Dr. Changchun Zhou](#)

- Designed a **GEMV-centric Compute Engine** tailored for transformer inference, exploiting weight-stationary dataflow and hierarchical tiling to alleviate memory bandwidth bottlenecks. Incorporated structured sparsity and INT4/INT8 quantization to improve compute density and energy efficiency.
- Developed a **VQ-based Compression and Retrieval Module** that encodes activations/weights into compact codebooks, enabling efficient vector lookup and reducing DRAM access. Integrated approximate nearest neighbor (ANN) search with hardware-friendly distance computation, leading to substantial savings in memory footprint and latency for LLM inference.
- Proposed a **GEMV–VQ Co-design Framework** that fuses vector quantization with matrix-vector multiplication, allowing direct computation in compressed domain and minimizing dequantization overhead, improving throughput and reducing end-to-end energy consumption.
- Led **NPU architecture design** for LLM acceleration, defining microarchitecture, memory hierarchy, and interconnect across compute modules. Coordinated cross-module optimization to balance compute utilization and memory bandwidth under tight power constraints.
- Directed RTL development and design space exploration, analyzing trade-offs among precision, throughput, and silicon cost; achieved scalable architecture supporting variable sequence lengths and model sizes.

RISC-V 3-Stage Pipelined CPU with UART on FPGA

Mar 2025 - May 2025

UC Berkeley, USA

Project leader | Instructed by Prof. John Wawrzynek

- Designed and implemented a three-stage pipelined RISC-V CPU with complete I/O peripheral circuits, including FIFO, UART and related modules in Verilog, targeting the Xilinx PYNQ-Z1 platform; achieved full RV32I instruction set support.
- Built all pipeline components from scratch including hazard detection unit, forwarding logic, and branch prediction logic, reducing CPI from 1.60 to 1.12 on a matrix multiplication benchmark.
- Created testbenches to validate CPU and I/O modules at both unit and system levels.
- Optimized critical path and control logic to increase maximum frequency from 50 MHz to 90 MHz (top 10% in class) and reduced LUT-based area from 200,000 to 140,000.

OTHER PROJECTS

Analog Front-End Circuit Design for Mixed-Signal IoT Chip

Mar 2025 - May 2025

Image Recognition in Artificial Intelligence Fundamentals

Nov 2024 - Jan 2025

Thermal Dissipation Mechanism of High-Temperature Hydrogels

Jun 2024 - Jan 2025

SKILLS

Digital IC Design: Front-end SoC IP design for AI accelerators, including RTL coding, verification, and DPI-C co-simulation; familiar with AXI bus protocol.

Tools & EDA Software: Cadence, Vivado, ModelSim, VCS.

Programming Languages: Verilog & SystemVerilog, Python, C/C++, SQL, Matlab.

Languages: English: Fluent, TOEFL iBT 106(Speaking 25, Writing 27); Chinese(Mandarin): Native

EXTRA-CURRICULAR ACTIVITIES

Head of Wuhan University Student Art Troupe

Sep 2024 - Aug 2025

Invited to attend the 2024 WA Chinese New Year Ball and sing for Australian Prime Minister Albanese

Collaborated with Mr. Dai Yuqiang, one of the "Three Tenors of China"

Member of University Volunteer Center

Sep 2022 - Sep 2023

Member of University Chinese Kungfu Association

Sep 2022 - Sep 2023

Hobbies: Classical Tenor Singing, Ski, Tennis